# Assignment #01

Name : M. Shoaib Shoukat
Sap ID : 28913
Course : operating system

## Q#01

| Student | Arrival time | Burst time |
|---|---|---|
| A | 0 | 6 |
| B | 2 | 3 |
| C | 4 | 4 |
| D | 6 | 5 |

## 1. Round Robin Scheduling

∵ Time Quantum
∵ Preemptive

| St | A.T | B.T | C.T | TAT (CT−AT) | WT (TAT−BT) |
|---|---|---|---|---|---|
| A | 0 | ~~3~~ 3 | 12 | 12 | 6 |
| B | 2 | 3 | 6 | 4 | 1 |
| C | 4 | ~~4~~ 1 | 18 | 14 | 7 |
| D | 6 | ~~5~~ 2 | 18 | 12 | 7 |

time Quantum = 3

Ready Queue

| A | B | A | C | D | C | D |
|---|---|---|---|---|---|---|

Running Queue

| A | B | A | C | D | C | D |
|---|---|---|---|---|---|---|

0 3 6 9 12 15 16 18

## 2. SJF (Shortest Job First)

- Non-preemptive

| Student | A.T | B.T | C.T | TAT (CT-AT) | W.T (TAT-BT) |
|---|---|---|---|---|---|
| A | 0 | 6 | 6 | 6 | 0 |
| B | 2 | 3 | 9 | 7 | 4 |
| C | 4 | 4 | 13 | 9 | 5 |
| D | 6 | 5 | 18 | 12 | 7 |

Gantt-chart

| A | B | C | D |
|---|---|---|---|

0 6 9 13 18

## 3- FCFS (First Come- First Served)

- Non-preeptive scheduling

| Student | A.T | B.T | C.T | TAT (CT-AT) | W.T (TAT-BT) |
|---|---|---|---|---|---|
| A | 0 | 6 | 6 | 6 | 0 |
| B | 2 | 3 | 9 | 7 | 4 |
| C | 4 | 4 | 13 | 9 | 5 |
| D | 6 | 5 | 18 | 12 | 7 |

Gantt chart:

| A | B | C | D |
|---|---|---|---|

0 6 9 13 18

## Q#02

| Process | BoT | Priority | C.T | TAT (CT-AT) | W.T (TAT-BT) |
|---|---|---|---|---|---|
| P5 | 3 | 4 | 3 | 3 | 0 |
| P1 | 8 | 3 | 11 | 11 | 3 |
| P3 | 9 | 2 | 20 | 20 | 11 |
| P4 | 5 | 2 | 25 | 25 | 20 |
| P2 | 4 | 1 | 29 | 29 | 25 |

Gantt chart

| P5 | P1 | P3 | P4 | P2 |
|---|---|---|---|---|

0 3 11 20 25 29

$$\text{average time} = \frac{\Sigma \text{ waiting times}}{\text{No of process}} = \frac{59}{5}$$

$$59/5 = \boxed{11.8 \text{ ms}}$$

## Advantages

1. Priority scheduling ensures that high priority tasks are processed first, which is ideal in real-time systems where certain tasks must be completed as soon as possible.
2. Allows for processes with different levels of importance to be scheduled accordingly.

Disadvantages -

1. Low-priority tasks may suffer indefinite delays if higher-priority processes keep arriving, leading to starvation.

2. In systems with multiple dependencies a low-priority task may block a high priority task, leading to a situation known as priority inversion.

3. Determining and managing priority levels can be challenging and may require a dynamic approach to prevent low priority processes from waiting too long.

## Q#03

| Process | Burst time | Arrival time |
|---|---|---|
| P1 | 10 | 4 |
| P2 | 15 | 3 |
| P3 | 18 | 1 |
| P4 | 16 | 2 |
| P5 | 20 | 4 |

**A.**

| Process | A.T | B.T | C.T | TAT (CT-AT) | W.T (TAT-BT) |
|---|---|---|---|---|---|
| P3 | 1 | 18 | 19 | 18 | 0 |
| P1 | 4 | 10 | 29 | 25 | 15 |
| P4 | 2 | 16 | 45 | 43 | 27 |
| P2 | 3 | 15 | 60 | 57 | 42 |
| P5 | 4 | 20 | 80 | 78 | 56 |

average time = $140/5 = 28$

average turnaround time = $\frac{219}{5} = 43.8$

B.

| Process | B.T | A.T | C.T | TAT (CT-AT) | WT (TAT - BT) |
|---|---|---|---|---|---|
| P1 | 10 | 4 | 37 | 33 | 23 |
| P2 | 15 | 3 | 61 | 58 | 43 |
| P3 | 18 | 1 | 57 | 56 | 38 |
| P4 | 16 | 2 | 65 | 63 | 47 |
| P5 | 20 | 4 | 80 | 76 | 56 |
| | | | | | |

Gantt-Chart

| P3 | P2 | P4 | P1 | P5 | P3 | P2 | P4 | P1 | P5 | P3 | P2 | P4 | P5 | P3 | P2 | P4 | P5 | P3 | P5 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

0 5 9 13 17 21 25 29 33 37 41 45 49 53 57 61 65 69 73 77 80

average time = 41.4

average TAT = 57.2

Q#04

| Process | Priority | A.T | B.T |
|---|---|---|---|
| P1 | 3 | 0 | 8 |
| P2 | 4 | 1 | 2 |
| P3 | 4 | 3 | 4 |
| P4 | 5 | 4 | 1 |
| P5 | 2 | 5 | 6 |
| P6 | 6 | 6 | 5 |
| P7 | 1 | 10 | 1 |

1. Priority Scheduling

| Process | A-T | B-T | C.T | TAT | W.T |
|---|---|---|---|---|---|
| P1 | 0 | 8 | 22 | 22 | 14 |
| P2 | 1 | 2 | 13 | 12 | 10 |
| P3 | 3 | 4 | 17 | 14 | 10 |
| P4 | 4 | 1 | 18 | 14 | 13 |
| P5 | 5 | 6 | 9 | 4 | 2 |
| P6 | 6 | 5 | 27 | 21 | 16 |
| P7 | 10 | 1 | 28 | 18 | 17 |

Gantt-chart

| P1 | P5 | P1 | P2 | P3 | P4 | P1 | P6 | P7 |
|---|---|---|---|---|---|---|---|---|

0 1 9 11 13 17 18 22 27 28

→ average time = 68/7 = 9.71

→ average TAT = 108/7 = 15

## 2. Priority Scheduling (Non-Preemptive)

| Process | A.T | B.T | C.T | TAT | W.T |
|---|---|---|---|---|---|
| P1 | 0 | 8 | 8 | 8 | 0 |
| P2 | 1 | 2 | 10 | 9 | 7 |
| P3 | 3 | 4 | 14 | 11 | 7 |
| P4 | 4 | 1 | 15 | 11 | 10 |
| P5 | 5 | 6 | 21 | 16 | 10 |
| P6 | 6 | 5 | 27 | 21 | 16 |
| P7 | 10 | 1 | 28 | 18 | 17 |

Gantt - Chart

| P1 | P2 | P3 | P4 | P5 | P6 | P7 |
|---|---|---|---|---|---|---|

0 8 10 14 15 21 27 28

→ Average waiting time = 67/7 ≈ 9.57

→ Average TAT = 94/7 ≈ 13.43